Mathematical model: hybrid CSP + PVT + wind + biomass plant

Assignment 2, Questions 6–10. Extends the Assignment 1 model; CSP and PVT physics are unchanged.

1. Sets

Symbol	Definition
Days	{June, December}
Hrs	$\{0, 1, \ldots, 23\}$
$\text{SunHrs}_{\text{day}}$	$\{\text{hr} \in \text{Hrs} : \text{Irradiance}_{\text{hr,day}} > 0\}$, hours with sunlight
Sources	{CSP, PVT, wind, biomass}

2. Parameters

Data, for all hr ∈ Hrs, day ∈ Days:

Symbol	Meaning	Unit
$\text{Irradiance}_{\text{hr,day}}$, $\text{T}_{\text{ambient,hr,day}}$	solar irradiance, ambient temperature (Assignment 1 file)	W/m^2, °C
$\text{Speed}_{\text{wind,hr,day}}$	wind speed (Assignment 2 file, km/h ÷ 3.6)	m/s
$\text{Demand}_{\text{elec,hr,day}}$, $\text{Demand}_{\text{heat,hr,day}}$	electricity and heat demand in that hour	GJ

CSP and PVT: all values of Assignment 1 (reflectivity 0.95, cosine 0.86, shading and blocking 0.89, atmospheric 0.97, spillage 0.92, receiver 0.80, $\text{Eff}_{\text{rankine}} = 0.42$, 220 m^2 per heliostat; PVT: 0.16, 0.86, 12 $\text{W/(m}^2\text{K)}$, 0.65, 35 °C, 4 m^2 per panel; $\text{Area}_{\text{max}} = 450\,000$ m^2; $\text{GJ}_{\text{per W-hr}} = 3.6 \times 10^{-6}$).

Wind

Symbol	Meaning	Value
$\text{Limit}_{\text{Betz}}$	maximum power coefficient of any turbine	$16/27 = 0.593$
$\text{Frac}_{\text{of Betz}}$	turbine efficiency as a fraction of the Betz limit	0.85 (Q10: 0.75–0.95)
$\text{Radius}_{\text{rotor}}$	rotor radius	36.5 m
$\text{Density}_{\text{air}}$	air density, sea level (Lecture 6; assumed)	1.225 kg/m^3

Biomass (ethanol)

Symbol	Meaning	Value
$\text{HHV}_{\text{ethanol}}$	higher heating value	29.7 MJ/kg
$\text{H}_{\text{evap water}}$	heat of evaporation of water	2.5 MJ/kg
$\text{Mass}_{\text{water per kg ethanol}}$	water formed per kg ethanol, $3 \times 18.015/46.07$ (Q2)	1.173 kg/kg

Rules

Symbol	Meaning	Value
$\text{Share}_{\text{CSP,min}}$	minimum CSP share of all electricity produced in the day (surplus included)	0.60
$\text{Weight}_{\text{unmet}}$	penalty per GJ of unmet demand ($\gg 1$ per GJ of surplus)	1000

3. Derived parameters

Energy one unit can deliver in each hour [GJ], for all hr, day:

$$\begin{aligned}
\text{Energy}_{\text{per heliostat,hr,day}} &= \text{GJ}_{\text{per W-hr}} \cdot \text{Area}_{\text{per heliostat}} \cdot \text{Eff}_{\text{CSP}} \cdot \text{Irradiance}_{\text{hr,day}} \\
\text{Energy}_{\text{per PVT elec,hr,day}} &= \text{GJ}_{\text{per W-hr}} \cdot \text{Area}_{\text{per PVT panel}} \cdot \text{Eff}_{\text{PVT,elec}} \cdot \text{Irradiance}_{\text{hr,day}} \\
\text{Energy}_{\text{per PVT heat,hr,day}} &= \text{GJ}_{\text{per W-hr}} \cdot \text{Area}_{\text{per PVT panel}} \cdot \text{Yield}_{\text{PVT heat,hr,day}} \qquad \text{(Assignment 1)} \\
\text{Energy}_{\text{per turbine,hr,day}} &= \text{GJ}_{\text{per W-hr}} \cdot \tfrac{1}{2}\,\text{Density}_{\text{air}} \cdot \pi\,\text{Radius}^2_{\text{rotor}} \cdot \text{Speed}^3_{\text{wind,hr,day}} \cdot \text{Coef}_{\text{power}}
\end{aligned}$$

with $\text{Eff}_{\text{CSP}} = 0.218$ from Assignment 1 and $\text{Coef}_{\text{power}} = \text{Frac}_{\text{of Betz}} \cdot \text{Limit}_{\text{Betz}} = 0.504$.

Ethanol. The flue gas leaves as vapour, so the heat delivered per kg is the lower heating value (as in Q5):

$$\text{LHV}_{\text{ethanol}} = \text{HHV}_{\text{ethanol}} - \text{Mass}_{\text{water per kg ethanol}} \cdot \text{H}_{\text{evap water}} = 26.77 \text{ MJ/kg}$$

4. Decision variables

Symbol	Meaning	Domain
$\text{Count}_{\text{heliostats}}$, $\text{Count}_{\text{PVT panels}}$, $\text{Count}_{\text{turbines}}$	number of units installed	$\{0, 1, 2, \dots\}$
$\text{Use}_{\text{s,hr,day}}$, s ∈ Sources	electricity source s delivers to demand in that hour	≥ 0 [GJ]
$\text{Unmet}_{\text{hr,day}}$	electricity demand not met in that hour	≥ 0 [GJ]

Energy *available* from the installed units in each hour [GJ]:

$$\begin{aligned}
\text{Avail}_{\text{CSP,hr,day}} &= \text{Count}_{\text{heliostats}} \cdot \text{Energy}_{\text{per heliostat,hr,day}} \\
\text{Avail}_{\text{PVT,hr,day}} &= \text{Count}_{\text{PVT panels}} \cdot \text{Energy}_{\text{per PVT elec,hr,day}} \\
\text{Avail}_{\text{wind,hr,day}} &= \text{Count}_{\text{turbines}} \cdot \text{Energy}_{\text{per turbine,hr,day}}
\end{aligned}$$

5. Full model

Objective: meet as much demand as possible, then create as little surplus as possible.

$$\min \sum_{\text{day}} \sum_{\text{hr}} \text{Weight}_{\text{unmet}} \cdot \text{Unmet}_{\text{hr,day}} \;+\; \text{Surplus}_{\text{elec}} \;+\; \text{Surplus}_{\text{heat}}$$

$$\text{Surplus}_{\text{elec}} = \sum_{\text{day}} \sum_{\text{hr}} \sum_{\text{s}\in\{\text{CSP,PVT,wind}\}} \left(\text{Avail}_{\text{s,hr,day}} - \text{Use}_{\text{s,hr,day}}\right) \qquad \text{(generated but not used)}$$

$$\text{Surplus}_{\text{heat}} = \sum_{\text{day}} \Big(\sum_{\text{hr}\in\text{SunHrs}_{\text{day}}} \text{Count}_{\text{PVT panels}} \cdot \text{Energy}_{\text{per PVT heat,hr,day}} - \sum_{\text{hr}\in\text{SunHrs}_{\text{day}}} \text{Demand}_{\text{heat,hr,day}} \Big)$$

Ethanol is burned only when needed, so it never creates surplus.

Dispatch order. When several renewables could serve the same hour, which one is curtailed does not change the total surplus. A negligible weight (10^{-4} per GJ) fixes the order: wind first (preference to wind), then CSP, then PVT (built for heat, so its electricity is the most expendable); ethanol carries 10^{-3} per GJ so it is used only when nothing else can serve demand.

Constraints, for all day ∈ Days:

Heat demand over the sunlit hours, unchanged from Assignment 1 (PVT is the only heat source):

$$\sum_{\text{hr}\in\text{SunHrs}_{\text{day}}} \text{Count}_{\text{PVT panels}} \cdot \text{Energy}_{\text{per PVT heat,hr,day}} \;\geq\; \sum_{\text{hr}\in\text{SunHrs}_{\text{day}}} \text{Demand}_{\text{heat,hr,day}} \tag{C1}$$

Electricity balance in *every* hour, for all hr $\in$ Hrs:

$$\text{Use}_{\text{CSP,hr,day}} + \text{Use}_{\text{PVT,hr,day}} + \text{Use}_{\text{wind,hr,day}} + \text{Use}_{\text{biomass,hr,day}} + \text{Unmet}_{\text{hr,day}} = \text{Demand}_{\text{elec,hr,day}} \tag{C2}$$

A source cannot deliver more than it generates, for all hr and s $\in$ {CSP,PVT,wind}:

$$\text{Use}_{\text{s,hr,day}} \leq \text{Avail}_{\text{s,hr,day}} \tag{C3}$$

CSP produces at least 60% of *all* electricity produced in the day, surplus included. Ethanol produces only what it delivers, so its production is $\text{Use}_{\text{biomass}}$:

$$\sum_{\text{hr}\in\text{Hrs}} \text{Avail}_{\text{CSP,hr,day}} \;\geq\; \text{Share}_{\text{CSP,min}} \sum_{\text{hr}\in\text{Hrs}} \left(\text{Avail}_{\text{CSP,hr,day}} + \text{Avail}_{\text{PVT,hr,day}} + \text{Avail}_{\text{wind,hr,day}} + \text{Use}_{\text{biomass,hr,day}}\right) \tag{C4}$$

Preference to wind: over the day, wind delivers at least as much electricity as ethanol (the brief: "while deciding on the share of wind and biomass, preference should be given to the wind turbine"):

$$\sum_{\text{hr}\in\text{Hrs}} \text{Use}_{\text{wind,hr,day}} \;\geq\; \sum_{\text{hr}\in\text{Hrs}} \text{Use}_{\text{biomass,hr,day}} \tag{C5}$$

Site area (collectors only; the turbines stand outside the solar site):

$$\text{Area}_{\text{per heliostat}} \cdot \text{Count}_{\text{heliostats}} + \text{Area}_{\text{per PVT panel}} \cdot \text{Count}_{\text{PVT panels}} \leq \text{Area}_{\text{max}} \tag{C6}$$

Whole units and non-negativity:

$$\text{Count}_{\text{heliostats}},\ \text{Count}_{\text{PVT panels}},\ \text{Count}_{\text{turbines}} \in \{0, 1, 2, \dots\}; \qquad \text{Use},\ \text{Unmet} \geq 0 \tag{C7}$$

This is a mixed-integer linear program (three integer variables, 240 continuous ones), solved with HiGHS via `scipy.optimize.milp`.

6. Results derived from the solution

Rate of ethanol combustion (Q7), for all hr, day [kg per hour]:

$$\text{Rate}_{\text{ethanol,hr,day}} = \frac{1000 \cdot \text{Use}_{\text{biomass,hr,day}}}{\text{Eff}_{\text{rankine}} \cdot \text{LHV}_{\text{ethanol}}} \qquad (\text{GJ} \to \text{MJ};\ \text{divide by } 3600 \text{ for kg/s})$$

Wind efficiency (Q10): $\text{Frac}_{\text{of Betz}}$ is varied from 0.75 to 0.95. (a) With the design fixed, the total electricity output is $\sum_{\text{hr}} \left(\text{Avail}_{\text{CSP}} + \text{Avail}_{\text{PVT}} + \text{Avail}_{\text{wind}} + \text{Use}_{\text{biomass}}\right)$, with (C4) and (C5) relaxed: the units are fixed, so ethanol simply fills whatever wind does not cover. (b) The model is re-solved at each value to give the number of turbines required.

Wind–ethanol trade-off (results workbook): $\text{Count}_{\text{turbines}}$ is fixed at $0, 2, \dots, 80$ and the rest of the model is solved without (C5). This shows how much surplus each extra turbine adds and how much ethanol it replaces; the design is the smallest count at which (C5) holds on both days.